SIMATS ENGINEERING
CO3- ASSESSMENT TOOL 2
SIMULATION TASK

COURSE NAME: DIGITAL SIGNAL PROCESSING   COURSE CODE: ECA0908

COURSE OUTCOME COVERED

CO 3: Evaluate finite word-length effects and multirate signal processing techniques, including decimation, interpolation, and polyphase decomposition. **Bloom Level mapped- BL3,BL4,BL5**

Topics Covered:
Quantization- Truncation and Rounding errors - Quantization noise - Limit cycle oscillations – Signal scaling.Parametric and Non parametric Spectral estimation. Decimation-Interpolation-Sampling rate conversion-Implementation of sampling rate conversion-Polyohase Decomposition -5G Decimation/Interpolation: For mmWave beamforming.

Assessment Tool number : **CO3-AT2**
Assessment Tool Weightage : **35%**
Assessment Tool Name : **Simulation Task**
Duration of this assessment : **60 Minutes**
Marks : **25 Marks**
Type of Assessment Conduction : **Self Learning (Home Task)**

Code of Conduct:

I **G.LAXMI MANIKANTA(192412651)**certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Question No	Conceptual Understanding 20	Model Design 25	Tool Usage 20	Analysis of Results 20	Documentation 10

Aim

To simulate a fixed-point FIR filter with and without signal scaling and evaluate the effectiveness of scaling in preventing arithmetic overflow.

Given Data

- FIR Filter Length = **12 taps**
- Sum of FIR Coefficients = **2.5**
- Input Amplitude = **±1**
- Word Length = **8 bits**

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1. Determine the Required Scaling Factor

Maximum output of the FIR filter is

$$Y_{max} = X_{max} \times \sum h\,(n)$$

Given,

$$X_{max} = 1$$
$$\sum h\,(n) = 2.5$$

Therefore,

$$Y_{max} = 1 \times 2.5 = 2.5$$

An 8-bit signed fixed-point number (normalized Q1.7 format) can represent approximately

$$-1 \le x < 1$$

Since

$$2.5 > 1$$

overflow will occur.

Hence, the required scaling factor is

$$\boxed{\text{Scaling Factor} = \frac{1}{2.5} = 0.4}$$

Scaled input amplitude

$$1 \times 0.4 = 0.4$$

Maximum output

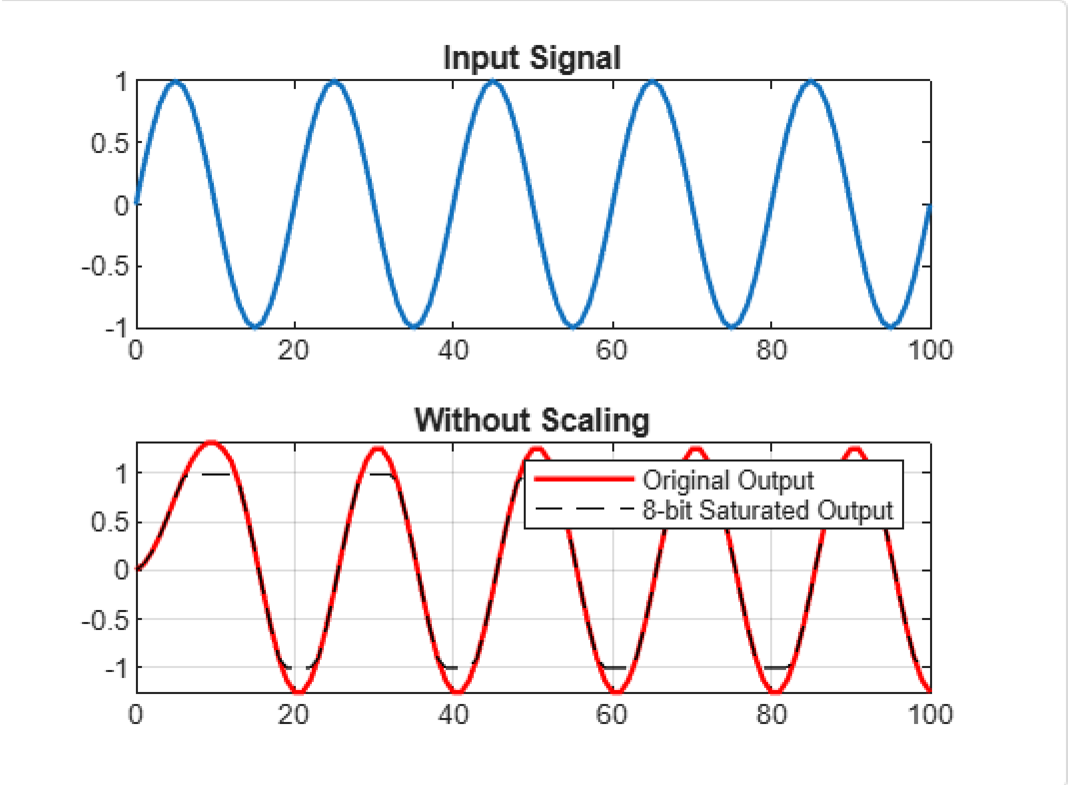
$$0.4 \times 2.5 = 1$$

Thus overflow is avoided.

Without Scaling

CODE :-

```
clc;
clear;
close all;

%% FIR Filter
N = 12;

% Coefficients having total sum = 2.5
h = ones(1,N)*(2.5/N);

%% Input Signal
n = 0:100;
x = sin(2*pi*0.05*n);

%% FIR Filtering
y1 = filter(h,1,x);

%% 8-bit Saturation
y1_sat = max(min(y1,0.992),-1);

%% Plot
figure;

subplot(2,1,1)
plot(n,x,'LineWidth',1.5)
title('Input Signal')

subplot(2,1,2)
plot(n,y1,'r','LineWidth',1.5)
hold on
plot(n,y1_sat,'k--')
title('Without Scaling')
legend('Original Output','8-bit Saturated Output')
grid on
```

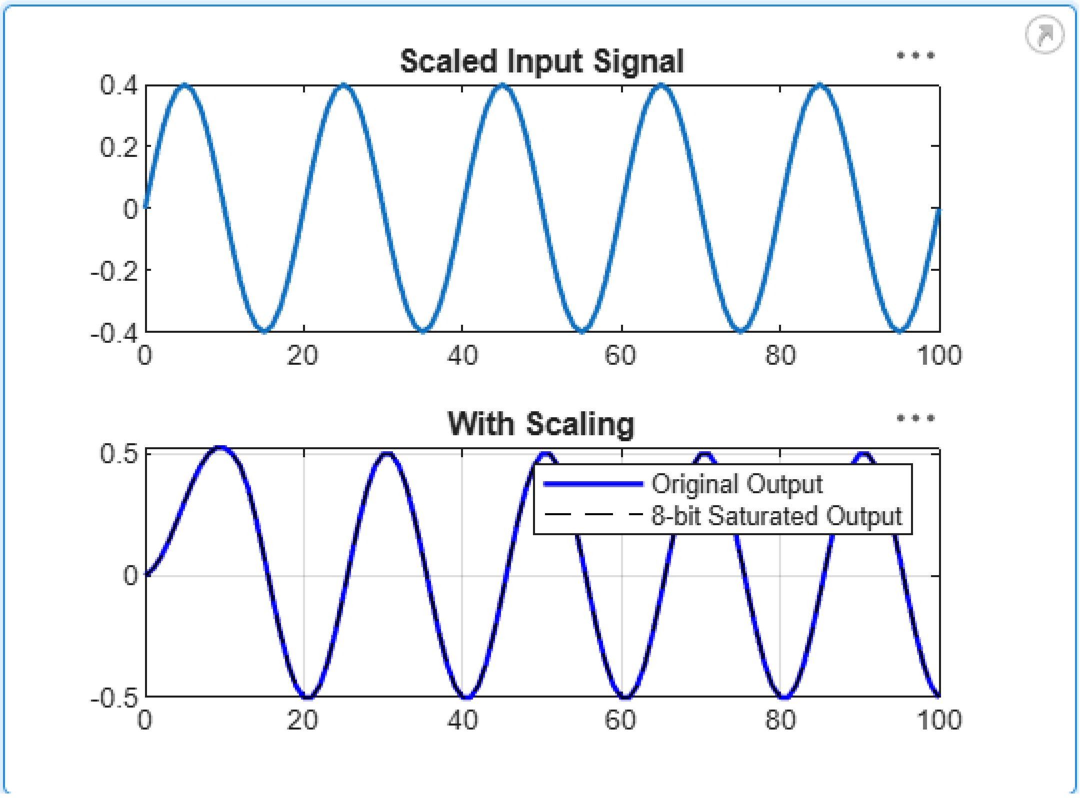


With Scaling :-

CODE :-

```
clc;
clear;
close all;

%% FIR Filter
N = 12;
h = ones(1,N)*(2.5/N);

%% Scaling Factor
SF = 1/2.5;

%% Input Signal
n = 0:100;
x = sin(2*pi*0.05*n);

%% Scale Input
x_scaled = SF*x;

%% FIR Output
y2 = filter(h,1,x_scaled);

%% Saturation
```

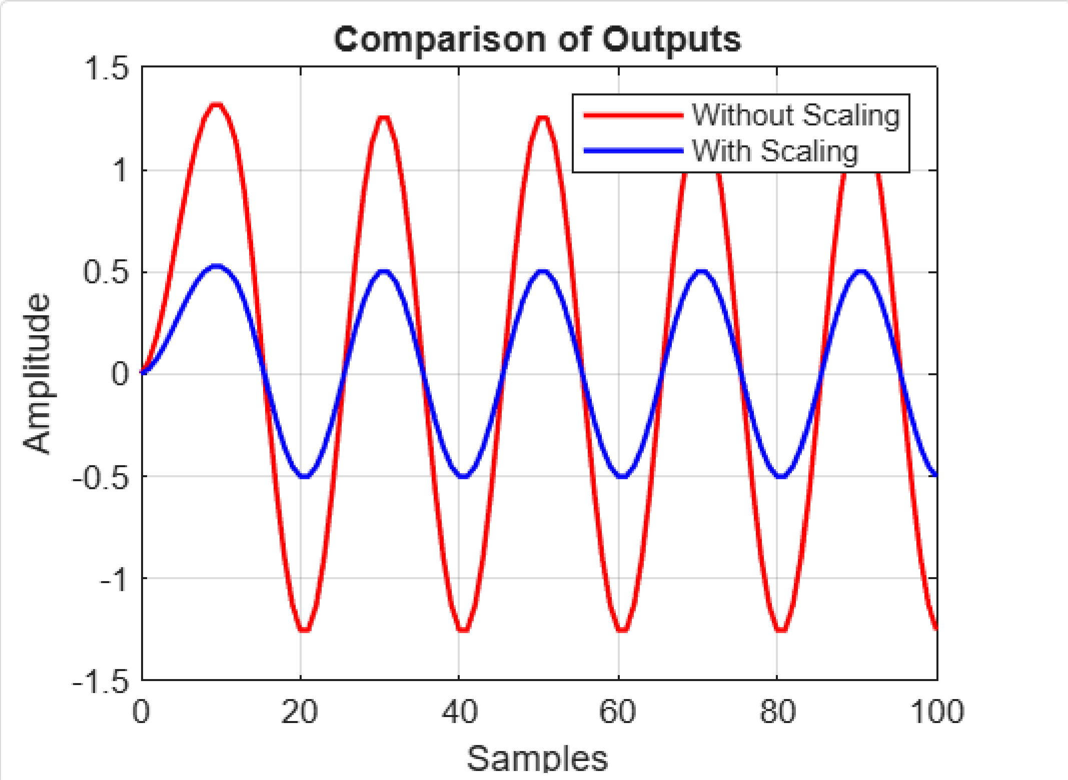
```
y2_sat = max(min(y2,0.992),-1);

%% Plot
figure;

subplot(2,1,1)
plot(n,x_scaled,'LineWidth',1.5)
title('Scaled Input Signal')

subplot(2,1,2)
plot(n,y2,'b','LineWidth',1.5)
hold on
plot(n,y2_sat,'k--')
title('With Scaling')
legend('Original Output','8-bit Saturated Output')
```



Compare Output Waveforms :-

CODE :-

```
clc;
clear;
close all;

N=12;
h=ones(1,N)*(2.5/N);

n=0:100;
x=sin(2*pi*0.05*n);

SF=1/2.5;

y1=filter(h,1,x);
y2=filter(h,1,SF*x);

figure

plot(n,y1,'r','LineWidth',1.5)
hold on
plot(n,y2,'b','LineWidth',1.5)

legend('Without Scaling','With Scaling')
title('Comparison of Outputs')
xlabel('Samples')
ylabel('Amplitude')
grid on
```



Identify Overflow Occurrences :-

CODE :-

```
overflow = abs(y1) > 1;

disp('Overflow Sample Numbers:')
disp(find(overflow))
```

```
Overflow Sample Numbers:

     8     9    10    11    12    13    20    21    22    23
```

6. Demonstrate the Effectiveness of Scaling

Parameter	Without Scaling	With Scaling
Maximum Input	±1	±0.4
Maximum Output	±2.5	±1
Overflow	Yes	No
Signal Distortion	High	None
Saturation	Present	Absent
Fixed-point Safe	No	Yes

Result :-

- **Required Scaling Factor: 0.4**
- Without scaling, the FIR filter output reaches approximately **±2.5**, exceeding the 8-bit fixed-point range and causing arithmetic overflow and saturation.
- With input scaling by **0.4**, the output remains within the allowable range (approximately **±1**), eliminating overflow while preserving the waveform shape.
- Signal scaling is therefore an effective technique for preventing arithmetic overflow in fixed-point FIR filter implementations.